

Modern Physics Notes

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Abstract

This is a simple abstract for the Modern Physics notes.

1 Introduction

Class test: 10%*2 (special relativity + quantum mechanics) + 80% exam (4 short(2+2) + 2 long)

2 Galileo Relativity

2.1 reference frame

It's about Newton's first law: for a reference frame that undergoes uniform linear motion or remains at rest, we call it an inertial reference frame. And clearly that for a reference frame that move relatively to a inertial reference frame in a constant speed, this frame is also inertial reference frame.

- Inertial reference frame: a reference frame in which Newton's first law holds.
- Non-inertial reference frame: a reference frame in which Newton's first law does not hold.

It is really important because when we use different reference frames to describe the motion of the same object, we may get different results.

2.2 velocity and acceleration

velocity is the rate of change of position with respect to time, and acceleration is the rate of change of velocity with respect to time.

$$\begin{aligned} \text{velocity} \quad v &= \frac{dx}{dt} \\ \text{acceleration} \quad a &= \frac{dv}{dt} = \frac{d^2x}{dt^2} \end{aligned}$$

for different theory, the definition of velocity and acceleration remain the same, because the definition of them is quite simple, but for the energy and motion, the definition is quite different.

2.3 time

there is only one thing to mention, in galileo relativity, the time is absolute, so velocity and acceleration depends on change in position vector, the only difference between different reference frame is different presentation of position vector. But in special relativity, the time changed.

2.4 Discribe motion in different reference frame

we define the position vector $\vec{r}$ and the velocity vector $\vec{v}$. it depends on the choose of the reference frame.

If we chose different reference frame, the position vector and the velocity vector will be different, related by the following equations:

$$\vec{r}' = \vec{r} + \vec{R} \quad (1)$$

(we asumed that the position vector of particle X in frame S is $\vec{r}$ and in frame S' is $\vec{r}'$, the subtract of the position vector of the origin of frame S' in frame S is $\vec{R}$)

Use differentiation, we can get the relation of the velocity vector in different reference frame, overall is same as the position vector

$$\vec{v}' = \frac{d\vec{r}'}{dt} \quad (2)$$

$$= \frac{d\vec{r}}{dt} + \frac{d\vec{R}}{dt} \quad (3)$$

$$= \vec{v} + \vec{V} \quad (4)$$

Remember the time in galileo relativity is absolute, $t' = t$, allow us to build this relationship, the relation of position vector is same with relation of velocity, as said "velocity and acceleration depends on change in position vector".